

System Operation Contract: createSupplierOrder

Operation:	createSupplierOrder(productID: Integer, initiator: String)
Cross References:	Use Cases: Create Supplier Order
Preconditions:	Product inventory is below minimum threshold (automatic trigger) OR Store Manager initiates manually. At least one supplier exists in the database for the product.
Postconditions:	- A SupplierOrder instance was created with attributes: orderID, supplierID, productID, quantity, orderDate, status='PENDING' (instance creation). - SupplierOrder was associated with the best available Supplier (association formed). - The order was sent to the Supplier System via the sendOrderRequest operation (operation called). - Order status was updated to 'SENT' upon confirmation (attribute modification). - If Supplier System rejected the order: status was set to 'FAILED' and an error alert was generated (attribute modification). - If supplier had limited quantity: a partial order was created and remaining quantity was noted (instance creation).